

Yifei Wang

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Ph.D. in Computer Science with industry experience in deep learning and AI for small-molecule drug discovery. Expertise spans virtual screening, generative molecular design, and molecular representation learning, drawing on graph neural networks, chemical language models, reinforcement learning, and efficient search. Architected and deployed production-level molecular design workflows, from billion-scale ligand-based virtual screening to synthesizable *de novo* design, working closely with computational and medicinal chemists to turn design constraints into robust ML algorithms that bridge advanced ML research and practical drug discovery.

RESEARCH TOPICS

Machine Learning, Deep Learning, Efficient Search, Optimization, Graph Neural Network, Generative AI, Reinforcement Learning, Transformer, Chemical Language Model, AI For Drug Discovery.

WORK EXPERIENCE

- Biogen** Massachusetts, USA
Scientist, Machine Learning & Molecular Design Mar 2026 - Present
Machine Learning Research Fellow Feb 2025 - Mar 2026
 - Focus on ultra-large virtual screening and synthesizable generative molecular design, integrating deep generative models and reinforcement learning within iterative "generate → evaluate → update" design loops.
 - Design agentic workflows in which LLM agents orchestrate molecular design tools end to end, planning and chaining generation, scoring, and filtering steps.
 - Deploy automated discovery pipelines for small molecules through cross-functional collaboration with computational and medicinal chemists, turning design constraints into tools used in internal projects.
- Brigham and Women's Hospital** Jul 2022 - Jan 2025
Research Trainee part-time Massachusetts, USA
 - Improved reliability and interpretability of healthcare AI models using EHR data across ML and LLMs.
 - Detected and evaluated fairness and bias in healthcare AI decision systems.
- XtalPi Inc.** Jul 2019 - Dec 2019
Algorithm Intern for AI-aided drug discovery Shenzhen, China
 - Implemented a transformer-like architecture for retrosynthetic reaction path prediction.
 - Improved model architectures and training strategies. Refined the pipeline of reaction prediction.

EDUCATION

- Brandeis University** Jan 2020 - Feb 2025
Ph.D. in Computer Science Massachusetts, USA
 - Research Focus: Deep Learning, Graph Neural Network, Molecular Representational Learning, Generative AI.
- Tsinghua University** Aug 2015 - Jul 2019
Bachelor of Pure and Applied Mathematics Beijing, China
 - Major Focus: Statistics, Efficient Algorithm Design, Optimization, Machine Learning.

SELECTED PUBLICATIONS

- [1] **Yifei Wang**, Nupur Bansal, Shiyun Wa, Simone Sciabola, Ye Wang. "Machine Learning-Aided Small-Molecule Virtual Screening: Recent Advances and Future Perspectives." *WIREs Computational Molecular Science*, 2026.
- [2] Shiyun Wa, **Yifei Wang**, Simone Sciabola, Ye Wang. "Advancing Ligand-based Virtual Screening and Molecular Generation with Pretrained Molecular Embedding Distance." In *ICML Workshop on AI for Science*, 2026.
- [3] **Yifei Wang***, Yunrui Li*, Lin Liu, Pengyu Hong, Hao Xu. "Advancing Drug Discovery with Enhanced Chemical Understanding via Asymmetric Contrastive Multimodal Learning." *Journal of Chemical Information and Modeling*, 2025.
- [4] **Yifei Wang**, Shiyang Chen, Guobin Chen, Ethan Shurberg, Hang Liu, and Pengyu Hong. "Motif-Based Graph Representation Learning with Application to Chemical Molecules." *Informatics*, vol. 10, no. 1, p. 8. MDPI, 2023.

*Equal contribution.

SELECTED PROJECTS

- **Project 1: Reinforcement-Learning-Driven Synthesizable Molecular Generative Design**

[\[Paper\]](#)[\[Code\]](#)

- Proposed a generative framework for ligand-based molecular design that solves the synthesizability problem by designing molecular structures together with their synthetic routes.
- Implemented goal-directed generation using on-policy reinforcement learning (PPO) to improve molecular structures while navigating multi-billion-scale chemical spaces.
- Adapted the framework to specialized molecular design requirements, supporting constraints that protect privileged groups and restrict generation to preferred reactions.

- **Project 2: Motif-based Graph Representation Learning on Molecules**

[\[Paper\]](#)[\[Code\]](#)

- Designed a novel convolution module for graph representational learning on molecules with an efficient pretraining strategy, enabling the capture of local structural and semantic information from graph motifs.
- Demonstrated the superiority of the proposed model in molecular learning across multiple dimensions: (1) better results in various molecular property prediction tasks, (2) more stable training, and (3) improved interpretability.

- **Project 3: Explainable Contrastive Multimodal Learning on Molecules**

[\[Paper\]](#)[\[Code\]](#)

- Proposed a training-efficient framework promoting cross-modality understanding between 2D molecular graphs and other chemical modalities.
- Demonstrated the effectiveness and interpretability of the proposed framework in various challenging tasks: (1) large-scale cross-modality retrieval, (2) isomer discrimination, and (3) ADMET prediction.

SKILLS

- **ML Frameworks:** PyTorch, Pytorch Geometric, Huggingface, Transformer Reinforcement Learning, Chemprop
- **Molecular Design:** Deep Reinforcement Learning (RL), Generative Models (Diffusion, GNNs, Transformers), Efficient Search (Thompson Sampling, Combinatorial Search).
- **Cheminformatics:** RDKit, SMILES/SMARTS Mapping, Virtual Screening.
- **Soft Skills:** Project and Time Management, Collaboration, Scientific Paper Writing, Mentorship.

SERVICES

- **Reviewer:** NeurIPS, ICLR, ICML, COLING, KDD Workshop, NIPS AI4Science Workshop, AMIA Annual Symposium, International Journal of Medical Informatics, BMC Medical Informatics and Decision Making
- **Teaching Assistant:** Computation Theory, Spring 2024 & Statistical Machine Learning, Fall 2020, 2021, 2022.
- **Undergraduate Thesis Advisor:** Supervised a thesis that earned the Kukin Moskowitz Prize for Excellence 2024.